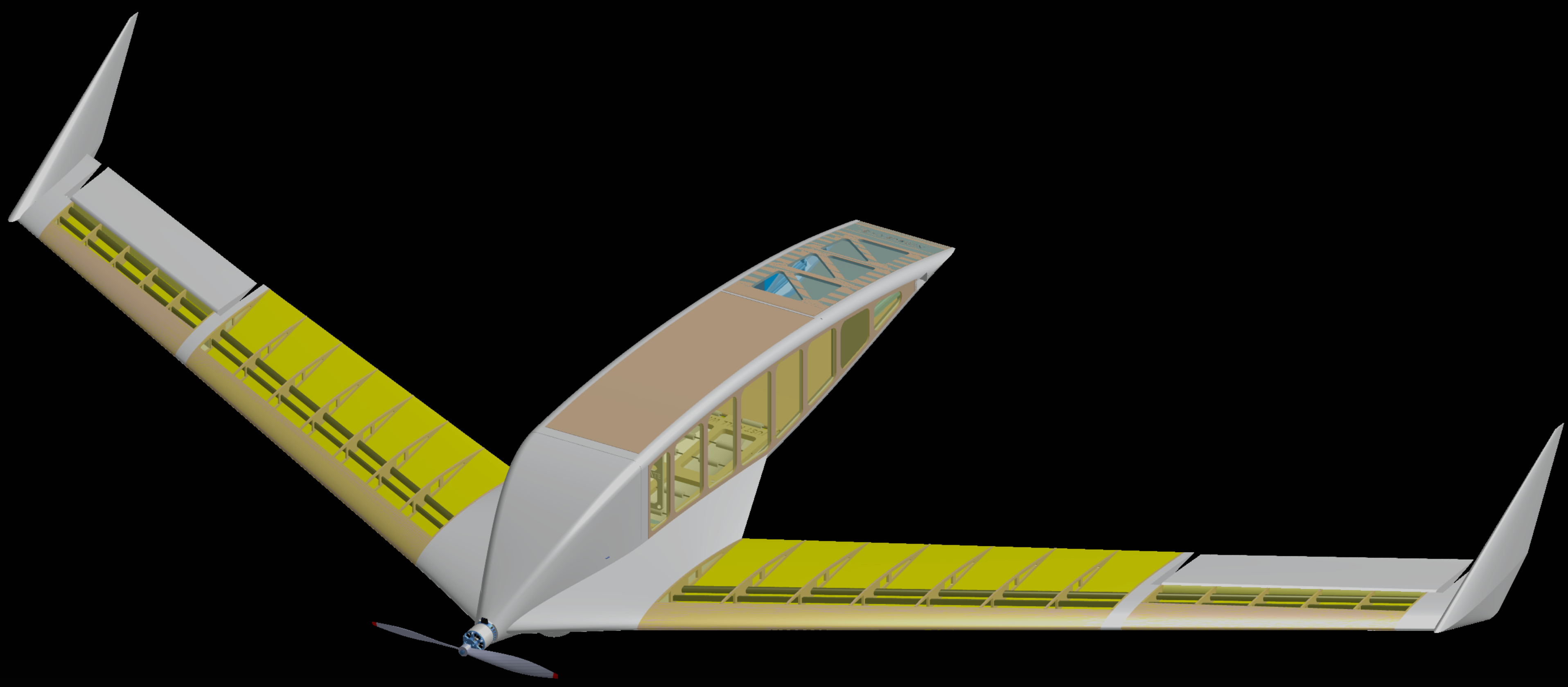
NIMBUS

PACKAGE DELIVERY UAV — SWEPT FLYING WING

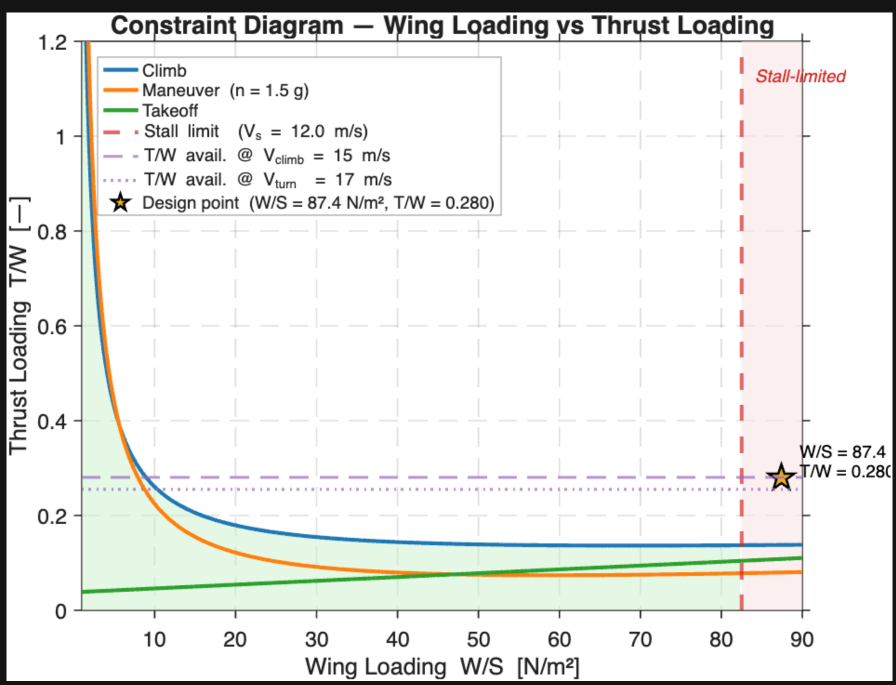
MAE 155B — Aircraft Design
UC San Diego · Spring 2026
6-Person Design Team



CMA-ES optimized · XFOIL surrogate aerodynamics · AVL dynamic stability · Flight tested Spring 2026

2.60 kg	20 m/s	14.34	800 g	$2.70/hr
MTOW	CRUISE SPEED	CRUISE L/D	PAYLOAD	PROFIT RATE

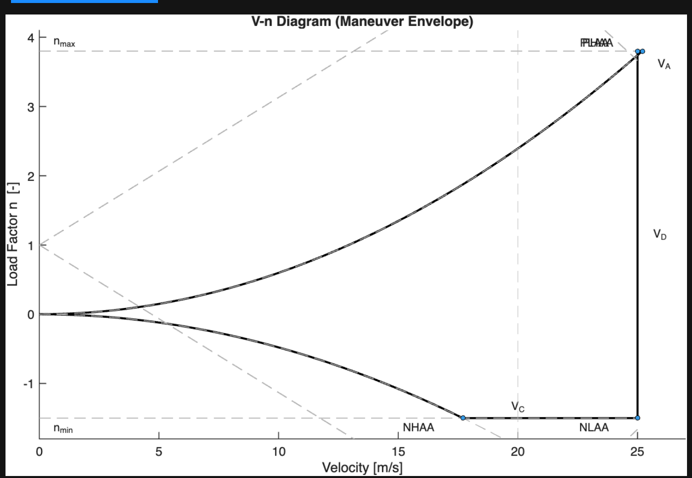
CTOL CONSTRAINT DIAGRAM



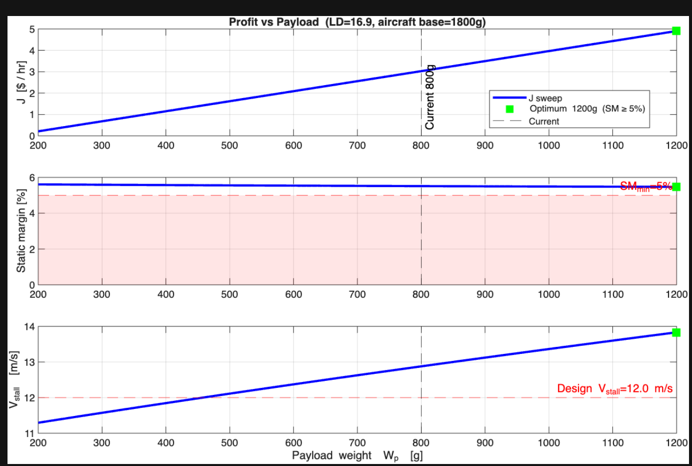
KEY DESIGN PARAMETERS

GEOMETRY		
Wingspan	**1.5715 m**	61.9 in
Wing area	**0.3087 m²**	—
Aspect ratio	**8.0**	—
Quarter-chord sweep	**28.3°**	—
Root / Tip airfoil	**E222 / E230**	—
Geometric washout	**–4.04°**	Panknin
WEIGHTS		
MTOW (loaded)	**2.60 kg**	—
Payload	**800 g**	—
PERFORMANCE		
Cruise speed	**20 m/s**	44.7 mph
Stall speed (loaded)	**13.3 m/s**	—
Wing loading W/S	**87.4 N/m²**	—
Cruise L/D	**14.34**	—
Static thrust	**10.2 N**	bench-corrected
PROPULSION		
Motor	**SunnySky X2216 V3**	1100 KV
Propeller	**APC 10×4.7SF**	—
Battery	**3S LiPo, 11.1 V**	—
STABILITY		
Static margin (AVL)	**4.88%**	MAC
Short-period ω■ / ζ	**6.68 / 0.62**	rad/s
Dutch roll ω■ / ζ	**4.24 / 0.08**	rad/s

V-n MANEUVER ENVELOPE



PROFIT vs PAYLOAD WEIGHT



AERODYNAMIC POLARS